

HUM 4441: ENGINEERING ETHICS — FINAL EXAM MASTER CHEAT SHEET

Core Reference Blueprint • 20 Canonical Case Studies • Core Ethical Theories • Decision Trees & Diagnostic Recipes • NSPE Canon 1 • Answering Scaffolds

EXAM ARCHITECTURE & MARKING BLUEPRINT

- **Rule:** Open-book exam; single book + **single stapled document**.
- **Format:** Exactly 4 compulsory questions × 30 marks = **120 marks total**. Time: 120 min → **1 mark per minute pace**.
- **OBE Mapping: PO6 (The Engineer and Society)** = 78 marks (65%); **PO8 (Ethics)** = 42 marks (35%). CO1: Safety/Problem Solving (43%), CO2: Professional Duty (24%), CO5: Code Hierarchy (16%), CO3: Issue Sorting (8%), CO4: Distributive Justice (8%).
- **Core Scoring Mandate:** Pure abstract moralizing without linking to engineering failure modes, code canons, or vulnerable populations triggers 40–60% deductions.

1. MASTER CASE STUDY MATRIX (20 CASES)

1. Citicorp Center (1978)

Actors: William LeMessurier (PE), Diane Hartley, Walter Wriston (Citibank CEO), Leslie Robertson.

Technical Dilemma: Stilt columns placed at center of faces; diagonal chevron wind bracing designed for perpendicular winds; contractor substituted welded joints with bolted joints without recalculating quartering (diagonal) winds; 40% increase in member stress, 160% in joints; 16-year storm failure threshold without Tuned Mass Damper (TMD); Hurricane Ella approaching NYC.

Ethical Violation: Initial failure to calculate quartering winds; press blackout during repairs concealed catastrophic collapse hazard from 200,000 residents.

Resolution: LeMessurier confessed to Citibank, insurers, building dept; 2-inch steel plates welded over 200+ bolted joints at night; \$2M insurance liability cap paid.

Takeaway: Construction substitutions (bolted vs welded) are design changes requiring full re-analysis. Public safety trumps professional reputation.

2. GE Sealed-Beam Headlamp (1930s)

Actors: Daniel Wright (GE lamp inventor), Val Roper, Corning Glass, Guide Lamp, C.M. Hall, Corcoran Brown, SAE.

Technical Dilemma: Unsealed headlamps tarnished, fogged, and corroded in months, cutting lighting by 50%+ and causing severe night highway fatalities.

Ethical Core: Positive ethics & supererogation. GE resisted monopolizing its patented sealed-beam design for corporate profit; openly shared tooling and cross-licensed to competitors.

Resolution: SAE adopted nationwide standard sealed-beam headlamp in 1940; night driving fatalities dropped dramatically across the United States.

Takeaway: Life-safety technology transcends commercial monopoly; engineering leadership standardizes public protections industry-wide.

3. GM Daytime Running Lamps (DRLs) (2001)

Actors: General Motors (GM), NHTSA, AIAM.

Technical Dilemma: DRLs reduce two-lane collision risk by 5–10% in northern latitudes (Canada, Scandinavia). GM petitioned NHTSA to mandate DRLs across all US vehicles.

Ethical Core: Rent-seeking disguised as safety advocacy. GM had already installed DRL production lines; lower US latitudes suffered headlight glare and motorcycle conspicuity loss.

Resolution: NHTSA denied GM's mandate petition in 2001, permitting DRLs as optional equipment only due to questionable net benefits.

Takeaway: Distinguish genuine universal safety mandates from corporate rent-seeking; safety claims must be empirically proven across diverse environments.

4. Automotive Crash Testing: NHTSA vs. IIHS

Actors: NHTSA (FMVSS 208 full-frontal rigid barrier at 35 mph), IIHS (Insurance Institute for Highway Safety, 40% moderate offset deformable barrier crash test at 40 mph).

Technical Dilemma: Automakers engineered cars to pass the legal minimum NHTSA full-frontal test by channeling loads through both longitudinal frame rails. Real-world crashes are offset, bypassing frame rails and crushing the passenger cabin.

Ethical Violation: Compliance as ethical floor. Automakers treated regulatory minimums as design ceilings, selling 5-star NHTSA cars that catastrophically collapsed in 40% offset crash tests.

Resolution: IIHS publicly shamed automakers with offset crash test videos; manufacturers redesigned unibody cabins with high-strength steel cages.

Takeaway: Legal compliance is the bare floor of engineering ethics, never the ceiling. Engineers must design for real physical crash dynamics, not test artifacts.

5. COMPAS Recidivism Algorithm (2016)

Actors: Northpointe (Equivant), ProPublica (Julia Angwin, Jeff Larson), Wisconsin Courts (*State v. Loomis*).

Technical Dilemma: 137-item questionnaire predicting 2-year recidivism. Black defendants suffered double the False Positive Rate (44.9% vs 23.5%); White defendants had double False Negative Rate (47.7% vs 28.0%).

Ethical Violation: Violates error rate parity. Trade Secret defense concealed proprietary scoring formula from criminal defendants, violating procedural due process.

Resolution: Courts allowed advisory use only with warning caveats; Kleinberg's Impossibility Theorem proved calibration and error rate parity cannot co-exist with unequal base rates.

Takeaway: Algorithmic fairness is mathematically constrained by base rates; proprietary opacity in sentencing violates fundamental human rights.

6. Amazon AI Recruitment Tool (2014–2018)

Actors: Amazon Machine Learning Team, HR Recruitment division.

Technical Dilemma: Resume scoring hiring tool and resume filter (1–5 stars) trained on 10 years of historical resumes submitted to Amazon (predominantly male engineers).

Ethical Violation: Historical & Representation bias. Model actively penalized resumes containing “women’s” (e.g. “women’s chess club”) and downgraded graduates of all-women’s colleges; picked up gendered proxy verbs.

Resolution: Project abandoned and disbanded in 2018; tool never deployed as sole hiring decider.

Takeaway: Machine learning automates and amplifies past societal discrimination. Removing explicit protected attributes fails when correlated proxy features remain.

7. Gender Shades (2018)

Actors: Joy Buolamwini (MIT Media Lab), Timnit Gebru (Stanford), IBM, Microsoft, Face++.

Technical Dilemma: Commercial facial analysis APIs achieved 99%+ accuracy on lighter males, but error rates reached 34.7% on darker females in Pilot Parliaments Benchmark.

Ethical Violation: Intersectional erasure, representation bias, and evaluation bias. Datasets were >75% lighter-skinned and >60% male; aggregate accuracy masked subgroup failures.

Resolution: IBM and Microsoft retrained models on diverse datasets; later suspended facial recognition sales to law enforcement.

Takeaway: Aggregate accuracy metrics mask catastrophic subgroup failure modes; algorithmic audits must evaluate intersectional demographic performance.

8. Obermeyer Healthcare Risk Tool (2019)

Actors: Ziad Obermeyer et al., commercial healthcare insurer tool affecting ~200M US patients.

Technical Dilemma: Algorithm predicted future healthcare costs (expenditure) as a proxy for healthcare need (health risk) to enroll patients in specialized care management.

Ethical Violation: Measurement bias. Systemic barriers mean less money is spent on Black patients; equal-cost Black patients were substantially sicker than White patients, denying care.

Resolution: Switching target variable from cost to actual health metrics (chronic conditions, lab results) increased Black patient enrollment from 17.7% to 46.5%.

Takeaway: Flawed proxy metrics embed structural inequities into algorithms; engineers must ensure proxy variables genuinely reflect latent health constructs.

9. YouTube Inciting Violence / Myanmar (2012–2018)

Actors: Google/YouTube (*Innocence of Muslims*), Meta/Facebook (Myanmar military propaganda against Rohingya), Cindy Lee Garcia, UN Human Rights Council.

Technical Dilemma: Engagement-maximizing recommender systems amplified inflammatory, dehumanizing hate speech 6.5x faster than truth, driving communal riots and ethnic cleansing.

Ethical Violation: Algorithmic amplification of lethal violence; causal distance defense (“we are neutral conduits”); failure to invest in local language content moderation (Burmese).

Resolution: Google temporarily geo-blocked video; UN declared Facebook played a “determining role” in Rohingya genocide; platforms faced global regulatory crackdowns.

Takeaway: Algorithmic curation is active editorial distribution, not neutral conveyance; platforms possess a moral duty of care when recommenders amplify violence.

10. Social Media Adolescent Mental Health

Actors: Meta (Instagram), Frances Haugen (whistleblower), Jonathan Haidt, US Senate Commerce Committee.

Technical Dilemma: Recommender systems optimized for engagement directed vulnerable teenagers into self-harm, eating disorder, and depression rabbit holes; adolescent self-harm and depression surged after 2010.

Ethical Violation: Corporate concealment. Meta’s internal research revealed 32% of teen girls felt Instagram worsened body image; prioritized ad revenue over child welfare.

Resolution: Frances Haugen leaked internal files (2021); ongoing state AG lawsuits; introduction of parental controls.

Takeaway: Kant’s Formula of Humanity forbids treating children as mere instruments of ad revenue; platforms owe heightened fiduciary duty to protect developing minds.

11. Therac-25 Radiation Overdoses (1985–1987)

Actors: AECL (Atomic Energy of Canada Limited), software developer, hospital operators, Ray Cox (victim).

Technical Dilemma: Linear accelerator replaced physical hardware interlocks with software control. Race condition during rapid operator keystrokes; cryptic “Malfunction 54”.

Ethical Violation: AECL dismissed hospital complaints, insisted machine was safe, blamed operators; delivered 25,000 rad overdoses (100x lethal dose) killing/injuring 6 patients.

Resolution: FDA recalled machine; AECL mandated hardware microswitch interlocks, software checksums, independent safety audits.

Takeaway: Never rely solely on software for life-critical safety; physical hardware interlocks are mandatory; investigate cryptic error codes with extreme urgency.

12. Space Shuttle Challenger (1986)

Actors: Roger Boisjoly, Arnie Thompson (Thiokol), Bob Lund (VP Engg), Jerald Mason (Thiokol GM), Larry Mulloy (NASA).

Technical Dilemma: O-ring rubber lost resiliency at sub-freezing launch temperatures (36°F ambient, 18°F joint), causing primary and secondary seal blowby.

Ethical Violation: Reversing burden of proof (“prove it will fail” vs “prove it is safe”); Mason ordered Lund: “Take off your engineering hat and put on your management hat”; NASA schedule pressure.

Resolution: Disintegrated 73s post-launch; 7 astronauts killed; Feynman ice-water demonstration; redesign of solid rocket joints.

Takeaway: Engineers must never subordinate professional safety judgment to management schedule pressure; lack of failure data does not prove safety.

13. Ford Pinto Fuel Tank Fires (1971–1978)

Actors: Ford Motor Company, Lee Iacocca, Harley Copp (whistleblower), NHTSA.

Technical Dilemma: Fuel tank placed 6 inches ahead of bumper behind rear axle without protective bladder; differential flange bolts punctured tank in ≥ 20 mph rear collisions.

Ethical Violation: Monetized cost-benefit analysis. Ford calculated fixing tank (\$11/car = \$137M) exceeded value of 180 burn deaths + 180 burn injuries (\$49.5M valuing life at \$200,000); delayed fix for 8 years.

Resolution: *Grimshaw v. Ford Motor Co.* landmark \$128M punitive damages; criminal indictment in Indiana; recall of 1.5 million Pintos.

Takeaway: Cost-benefit analysis cannot monetize human lives to justify marketing known lethal product flaws. Public safety trumps corporate profit.

14. Bhopal Chemical Disaster (1984)

Actors: Union Carbide Corporation (UCC), UCIL, Warren Anderson (CEO), Bhopal plant operators.

Technical Dilemma: Tank E610 stored 42 tons of methyl isocyanate (MIC); water entered during washing; exothermic runaway reaction blew relief valve; toxic cloud hit city.

Ethical Violation: Disabling safety safeguards to cut costs (\$30/day freon refrigeration turned off; vent gas scrubber disabled; flare tower unlit); double standards (India plant lacked US sister plant redundancies).

Resolution: >3,800 immediate deaths, >15,000 long-term deaths, >500,000 injured; UCC paid \$470M settlement; site remains toxic.

Takeaway: Safety standards are non-negotiable globally; cost-cutting that disables safety barriers in hazardous facilities is criminal negligence.

15. ValuJet Flight 592 (1996)

Actors: ValuJet Airlines, SabreTech (contractor), FAA, 110 victims.

Technical Dilemma: 144 expired chemical oxygen generators unthreaded without safety caps, wrapped in bubble wrap, labeled “empty canisters”, placed next to tires in DC-9 cargo hold lacking smoke/fire detection.

Ethical Violation: Systemic accident: subcontracted cost-cutting, lack of hazmat training, falsified maintenance logs (“ghost sign-offs”), non-redundant cargo hold fire suppression.

Resolution: Mid-air fire, crashed into Everglades at 500 mph; all 110 perished; SabreTech criminally convicted; FAA mandated fire suppression.

Takeaway: Safety depends on the entire socio-technical chain. If a design requires 100% human procedural perfection to avoid catastrophe, the engineering design is defective.

16. BART Automated Train Control (1970s)

Actors: Holger Hjortsvang, Max Blankenzee, Robert Bruder (BART engineers), BART Management, IEEE.

Technical Dilemma: World's first automated train control (ATC) transit system had flawed software, undetected ghost trains, and insufficient fail-safe braking logic.

Ethical Violation: Engineers raised warnings internally for a year; management fired all three for "insubordination" and blacklisted them; train later overshot Fremont station into sand-bank.

Resolution: IEEE filed historic *amicus curiae* brief supporting engineers; BART settled out of court; established whistleblowing protection precedents.

Takeaway: Public safety takes absolute precedence over employer loyalty under NSPE Canon 1; technical dissent regarding safety hazards is never insubordination.

17. Denver Airport Runway Concrete (1990s)

Actors: 3Bs Paving Contractor, Empire Laboratories (testing lab), City of Denver inspectors.

Technical Dilemma: Concrete mix for DIA runways diluted with excess water for easy pumping; failed compressive strength specifications and suffered alkali-silica reactivity (ASR) cracking.

Ethical Violation: Data falsification. Technicians altered computer break test reports, defending deliberate fabrication as "engineering judgment."

Resolution: Core drill samples revealed sub-standard runways; federal criminal prosecution; contractor executives convicted and sent to federal prison for wire/mail fraud.

Takeaway: "Engineering judgment" cannot alter or falsify empirical testing data; manipulating quality assurance records to hide non-conformance is criminal fraud.

18. Paradyne Computers RFP Fraud (1980)

Actors: Paradyne Corporation, Social Security Administration (SSA), former SSA procurement insider.

Technical Dilemma: SSA issued \$115M RFP requiring an existing, off-the-shelf distributed network computer system ("P8400").

Ethical Violation: Paradyne bid a non-existent system; bought competitor DEC PDP 11/23, pasted Paradyne labels over DEC logos, added dummy parts, and hired ex-SSA insider to rig the benchmark test.

Resolution: Whistleblower exposed fraud; SEC and DOJ filed civil and criminal charges; Paradyne paid millions in settlements.

Takeaway: Misrepresenting product maturity, rebadging competitor hardware, and exploiting revolving-door conflicts of interest corrupt competitive procurement.

19. Theranos Medical Diagnostic Fraud

Actors: Elizabeth Holmes, Sunny Balwani, Tyler Shultz, Erika Cheung, CMS, John Carreyrou.

Technical Dilemma: Claimed proprietary "Edison" finger-prick machine could run 200+ blood diagnostic tests from a single drop; microfluidic physics made tests wildly unreliable and inaccurate.

Ethical Violation: Secretly diluted patient samples and ran them on commercial Siemens machines; faked QC validation; issued false cancer/HIV/diabetes tests; silenced staff with NDAs.

Resolution: Whistleblowers Shultz and Cheung reported to CMS and WSJ; lab shut down; Holmes and Balwani convicted of criminal wire fraud, sentenced to prison.

Takeaway: "Fake it till you make it" is criminal fraud in medical/safety engineering; engineers have a moral obligation to whistleblow when false claims endanger health.

20. Coatzacoalcas Petrochemical Blast (2016)

Actors: PMV (Mexichem/Pemex joint venture), Clorados 3 VCM plant, Indigenous communities in Veracruz.

Technical Dilemma: Ruptured corroded pipe in vinyl chloride monomer (VCM) feed line released massive flammable gas cloud; exploded with TNT force; 32 killed, 136 injured, 2,000 evacuated.

Ethical Violation: Deferred maintenance (18-month inspection cycles due to oil price collapse cost-cutting); untrained inspectors lacking NDT tools; environmental racism near Indigenous communities.

Resolution: Government investigated; plant rebuilt under strict Safety Management System (SMS) oversight; demonstrated catastrophic cost of deferred maintenance.

Takeaway: Cost-cutting maintenance deferral on hazardous chemical lines guarantees catastrophe; environmental justice mandates equal safety protection for fence-line communities.

2. CORE ETHICAL THEORIES

Comparative Ethical Frameworks

- **Act Utilitarianism** (Bentham): Maximize net aggregate utility ($\Delta U = \sum B - \sum H$) for this specific act. **Pitfall:** **Tyranny of the Majority** (sacrifices vulnerable minorities; Ford Pinto).
- **Rule Utilitarianism** (Mill): Follow universal rules maximizing societal utility over time. Avoids ad-hoc corruption; supports engineering safety codes.
- **Kantian Deontology / Duty Ethics** (Kant):
 - **CI-1 (Universalizability)**: Can the maxim be willed as universal law without contradiction? (Model collapse, emissions test cheating).
 - **CI-2 (Formula of Humanity)**: Never treat persons merely as a means, but always as ends. (Autonomy, child data protection).**Pitfall:** Inflexible; struggles with conflicting absolute duties.
- **Rights Ethics** (Locke): Inalienable negative rights (non-injury, non-poisoning) act as trumps over collective utility. Negative rights bound cost-benefit optimization.
- **Virtue Ethics** (Aristotle): Ethics based on moral character, practical wisdom (*phronesis*), and **professional virtue**. **Aristotelian Golden Mean**: Courage is the mean between cowardice and recklessness.
- **Social Contract Theory** (Rawls): **Veil of Ignorance & Difference Principle**: Rules are just if agreed upon when personal status, race, and wealth are unknown.

3. DECISION TREES & QUICK RECIPES

3.1 Line-Drawing Technique

Purpose: Arbitrating borderline, ethically ambiguous cases.

Step 1: Define **Positive Paradigm (PP / A)** (100% ethical, fully transparent, safety-validated, informed consent).

Step 2: Define **Negative Paradigm (NP / G)** (100% unethical, fraudulent, reckless endangerment, active coverup).

Step 3: Select 4-6 evaluative features (Transparency, Severity of Harm, User Consent, Regulatory Compliance, Data Integrity).

Step 4: Assign weights (Public Safety features receive 3x-5x weight).

Step 5: Score test cases ($A \leftrightarrow G$ or $1 \leftrightarrow 10$). If weighted centroid falls closer to NP than PP, action is **impermissible**.

3.2 Accident Classification (3 Types)

- **Procedural Accident:** Operator error, checklist omission, failure to follow operational rules (e.g. Three Mile Island valve misread, pilot deviation).
- **Engineered Accident:** Flawed physical design, material failure, uncalculated stress load, edge cases (e.g. Citicorp bolted joint omission, Hyatt Regency hanger rod redesign).
- **Systemic Accident:** Complex interaction of organizational culture, cost pressures, fragmented subcontractors, and normalization of deviance (e.g. ValuJet 592, Challenger, Bhopal, Coatzacoalcas).

3.3 Whistleblowing: 4 Conditions (Harris/Pritchard/Rabins)

1. **Need:** Clear, serious, and imminent harm to the public, consumers, or workers (documented severe injury/death).
2. **Proximity:** Firsthand knowledge, direct observation, or official technical oversight (not rumor/hearsay).
3. **Capability:** Realistic competence, documentation, and authority to halt or mitigate the harm.
4. **Last Resort:** Internal escalation channels exhausted, or internal reporting is clearly futile, obstructed, or dangerous.

Permissible vs. Obligatory: Permissible when 1–4 met; **Morally Obligatory** when catastrophic loss of human life is certain without immediate external disclosure (NSPE Canon 1).

3.4 Conflict of Interest (COI) Diagnostic

- **Actual COI:** Personal or financial interests directly impair professional judgment or fiduciary duty. **Remedy:** Mandatory disclosure, recusal, and divestment.
- **Potential COI:** Interests do not currently impair judgment, but could reasonably do so as project evolves. **Remedy:** Formal disclosure, independent oversight firewall.
- **Apparent COI:** No actual bias, but a reasonable outside observer would reasonably doubt objectivity. **Remedy:** Disclosure, peer review, recusal if trust damaged.

3.5 The 7 Core AI Bias Types

1. **Historical Bias:** World data accurately reflects existing social discrimination (Amazon hiring tool).
2. **Representation Bias:** Certain demographic subgroups underrepresented in training set (Gender Shades darker females).
3. **Measurement Bias:** Flawed or noisy proxy features chosen to measure latent construct (Obermeyer healthcare cost vs illness).
4. **Aggregation Bias:** Monolithic model fails across distinct subpopulation dynamics (diabetes models).
5. **Evaluation Bias:** Benchmarks disproportionately represent majority group, masking subgroup errors.
6. **Deployment Bias:** Model used in context differing from training conditions (sentencing tool used for pre-trial cash bail).
7. **Feedback Loop Bias:** Predictions shape user actions that generate retrain data reinforcing initial bias (recommenders).

3.6 Fair Use 4-Factor Test (17 U.S.C. § 107)

1. **Purpose & Character:** Transformative use (adds new expression, meaning, or functional insight) vs commercial duplication.
2. **Nature of Work:** Factual/scientific (thinner copyright) vs highly creative/artistic/fictional (thick copyright).
3. **Amount & Substantiality:** Quantity used and qualitative “heart of the work”.
4. **Market Effect (Decisive Factor):** Does the use substitute for the original work or cannibalize its commercial licensing market?

3.7 IP Protection Selector

- **Copyright:** Original expression fixed in medium; automatic. Lasts author life + 70y (or 95y corp). Protects literal source code, object code, UI. No public disclosure required.
- **Patent (Utility):** Novel, non-obvious process/machine; overcomes *Alice* two-step. Lasts **20 Years**. Protects “faster way for computer to do X”, hardware optimizations. Mandatory public disclosure.
- **Trademark:** Distinctive marks identifying commercial source. Indefinite (10y renewable). Protects brand names, logos, distinctive product trade dress.
- **Trade Secret:** Secret info giving economic edge; reasonable security measures. Indefinite until leaked. Protects ML weights, training recipes, private search ranking algorithms.

4. NSPE CANON 1 VS LOYALTY

The Lexicographical Supremacy Principle

- **NSPE Canon 1:** “Hold paramount the safety, health, and welfare of the public.”
- **NSPE Canon 4:** “Act for each employer or client as faithful agents or trustees.”
- **Supremacy Rule:** Canon 1 is the supreme, non-negotiable trump card. An engineer’s duty to an employer is bounded (*prima facie*); an employer has no authority to order an engineer to compromise public safety.
- **Management Pressure:** Reject the “management hat” trap (Lund/Challenger). An engineer cannot shed safety responsibility by assuming a management title.
- **Dissent vs Insubordination:** Documented safety dissent is mandatory professional duty, never insubordination.

5. UNIVERSAL ANSWERING SCAFFOLDS

Scaffold 1: 2–4 Mark Questions (2–4 Min)

Structure (3 Sentences):

1. **Direct Answer:** Definitive Yes/No stating the primary ethical rule.
2. **Ethical Anchor:** Quote or define governing doctrine (e.g. Negative Rights, NSPE Canon 1, Procedural Accident).
3. **Case Link:** Apply directly to scenario facts with specific actors/numbers.

Scaffold 2: 5–8 Mark Questions (IRAC, 5–8 Min)

Structure:

- **Issue:** Frame the conflict (e.g. Canon 1 public safety vs Canon 4 deadline pressure).
- **Rule:** Define governing code canon, ethical theory test, or legal statute.
- **Application:** Apply rule systematically to facts; address counter-arguments (commercial cost, schedule pressure).
- **Conclusion:** State definitive moral verdict and actionable engineering mandate.

Scaffold 3: 10–12 Mark Questions (10–12 Min)

Option A: Six-Step Safety Design: 1. Define Problem (failure mode & root cause) → 2. Generate at least 2 distinct solutions (Hardware/Automation vs Institutional/Inspection) → 3. Multi-criteria analysis (reliability, cost, human error) → 4. Testing/simulation → 5. Select best hybrid → 6. Implementation & feedback.

Option B: Multi-Lens Comparison: Utilitarianism (ΔU , aggregate traps) vs Kantian Deontology (Universalizability, Mere Means) vs Rights Ethics (Negative inalienable rights). Select superior lens with justification.

Scaffold 4: 15 Mark Questions (15 Min)

- Line-Drawing Matrix:** 1. Define Positive Paradigm (PP / A) & Negative Paradigm (NP / G).
2. Construct 5-row table (Test Case, Action, Spectrum Position A–G, Score 1–10, Justification).
3. Identify distinguishing discriminative factors for each case.
4. Threshold synthesis: Draw boundary where commercial ambition crosses into criminal fraud and public endangerment.